BEING QUASI-SEPARATED IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'and we win' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Base-change stability and Zariski locality of quasi-separatedness. Schemes, Lemmas schemes-lemma-separated-permanence and schemes-lemma-characterize-quasi-separated, invoked at proof lines 5714-5718. Supplies the two preliminary halves of fpqc locality.
- Quasi-compactness descends along an fpqc covering. Same chapter, Lemma lemma-descending-property-quasi-compact, invoked at proof lines 5725-5727. Makes the diagonal of f quasi-compact.
- The general principle for a property that is fpqc local on the base. Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof line 5728. Concludes the locality statement.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KQ: sources/stacks-descent/pinned-original-source.tex, source lines 5688–5688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KO: sources/stacks-chow/tags.txt, source lines 2057–2057. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KU: sources/stacks-chow/tags.txt, source lines 2063–2063. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is quasi-separated" is fpqc local on the base.*

Proof. Any base change of a quasi-separated morphism is quasi-separated, see Schemes, Lemma 01KU. Being quasi-separated is Zariski local on the base (from the definition or by Schemes, Lemma 01KO). Finally, let $S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is quasi-separated. This means that $\Delta' : X' \to X' \times_{S'} X'$ is quasi-compact. Note that Δ' is the base change of $\Delta : X \to X \times_S X$ via $S' \to S$. By Lemma 02KQ this implies Δ is quasi-compact, and hence f is quasi-separated. Therefore Lemma 02KP applies and we win. □